Prime-field volcano layers, ecbench: RESULT

Two depth-8 ℓ=2 volcanoes over F_p (m0: p=40327618377097, m5: p=57628051254530177), 2 curves per interior level, 6 targets each, solvers rho.negation and rho.plain, counted unit ecbench.gae. Both sessions audited: 864 records, 864 verified, 8 exact replays reproduced, 0 problems. Host busy, isolation L0 (counts only; no wall time). Neighbour/vertex curves are not in the curve registry.

Measured

- m0: every ratio interval contains 1 (rho-neg floor/crater 0.889 [0.661, 1.255]; rho-plain 1.103 [0.822, 1.526]); level slope intervals contain 0.
- m5: rho-neg intervals all contain 1 (floor/crater 0.698 [0.498, 1.037]; slope −3.36% [−7.08, 2.09]). rho-plain middle/crater 0.688 [0.562, 0.891] and floor/crater 0.671 [0.524, 0.883] exclude 1; floor/middle 0.975 [0.793, 1.168] and the slope [−3.71, 5.53] do not.

Reading

Of the 16 tested intervals (4 ratios plus a slope, for each of 2 solvers, on 2 volcanoes), 14 contain the null value; the 2 that exclude it are both rho.plain on m5 and both involve the crater, which is one curve with 6 targets. This is 2 volcanoes; the crater is a single curve , so a crater-specific effect cannot be separated from a curve-specific one. Nothing here supports or refutes any transfer attack, since no degree-ℓ isogeny map was computed. The non-excluding intervals are wide (about ±30%) and do not show equality to within a small margin.

Tables

`m0`: p = 40327618377097, depth 8

rho-neg	middle / crater	0.784	[0.591, 1.102]	interval contains 1
rho-neg	floor / crater	0.889	[0.661, 1.255]	interval contains 1
rho-neg	floor / middle	1.135	[0.977, 1.336]	interval contains 1
rho-plain	middle / crater	0.972	[0.748, 1.325]	interval contains 1
rho-plain	floor / crater	1.103	[0.822, 1.526]	interval contains 1
rho-plain	floor / middle	1.134	[0.934, 1.375]	interval contains 1

`m5`: p = 57628051254530177, depth 8

rho-neg	middle / crater	0.824	[0.597, 1.209]	interval contains 1
rho-neg	floor / crater	0.698	[0.498, 1.037]	interval contains 1
rho-neg	floor / middle	0.847	[0.668, 1.072]	interval contains 1
rho-plain	middle / crater	0.688	[0.562, 0.891]	interval excludes 1
rho-plain	floor / crater	0.671	[0.524, 0.883]	interval excludes 1
rho-plain	floor / middle	0.975	[0.793, 1.168]	interval contains 1

Full per-level means: TABLES.md / tables.json. Figure: fig_levels.svg.

Koblitz n=83, ℓ=6473 (volcano_koblitz_n83_20261018)

Curve $y^2+xy=x^3+1$ over F_(2^83): t=-6151469093347, D_K=-7, f=6473·53676929, $\#E=2^2\cdot 241785163923079621668 5689$. Native koblitz-step built $\Phi_{6473}(X,j_0)$ from 9864 CRT primes (606365-bit coefficients, 8590 s total on 3 threads), reduced mod 2, factored into 78 Frobenius orbits of size 83, and found 6474 distinct level-1 j-invariants in F_(2^83) (expected 6474). 48 of 48 sampled neighbours had exactly one twist of order N; 0 of 8 random j passed the control. ALL CHECKS PASS. j-list and log hashes: SHA256SUMS. Not done: ecbench measurements at n=83 (the group is far outside rho reach) and no isogeny maps; the level-1 neighbours share N, so this verifies the volcano structure at ℓ=6473 only.

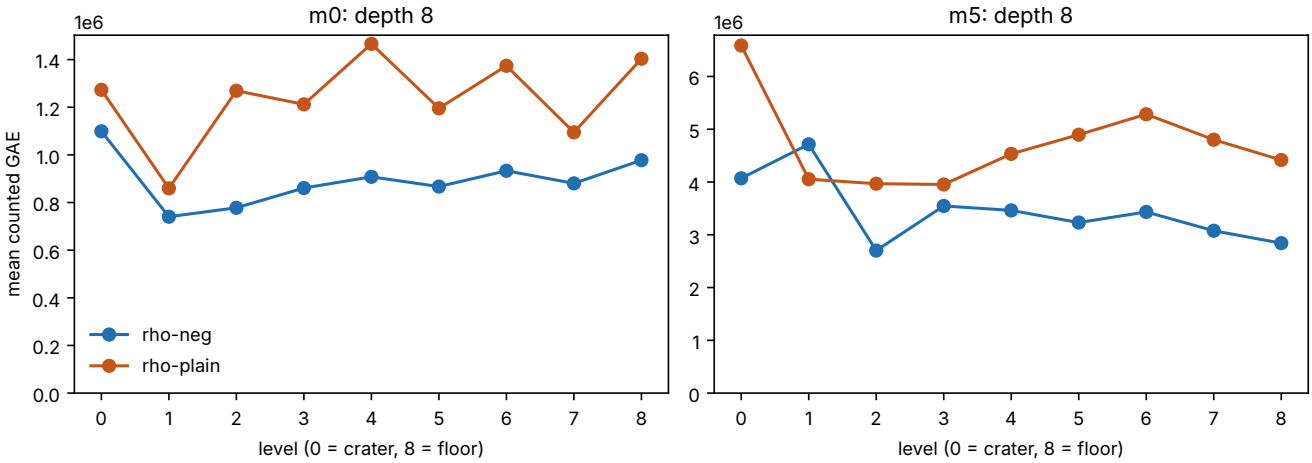

Figure 1: Mean counted GAE per level.